

WORKSHOP · 60 MINUTES · GITHUB REPO FOR THE CODE I WILL BE WALKING THROUGH TODAY →



AgentOps with MLflow

Practical MLflow Workflows for Monitoring,
Evaluation & Continuous Improvement

Stephanie Alba · Databricks

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What We'll Cover Today

A structured 60-minute workshop

#	Section	Time
1	Challenges of AgentOps	5 min
2	Five Pillars of AgentOps	5 min
3	Python Notebooks (Track A: Databricks / Track B: OSS)	40 min
4	Key Takeaways & Q&A	10 min



Most teams ship agents.

Few teams know how they're performing.

Without structured observability, you're flying blind —
relying on user complaints, spot checks, and gut feel to understand agent quality.

Today we fix that.

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Why Agents Are Uniquely Challenging

Traditional ML monitoring wasn't built for this

1 Non-Deterministic Outputs

The same input can produce different responses. You can't rely on exact-match checks -- quality is inherently distributional.

2 Multi-Step Execution Chains

Errors can occur anywhere in a pipeline. Retrieval failures, tool misuse, and hallucinations compound across steps.

3 Latency Variability

Token generation, tool calls, and retries create high p95/p99 tail latency. Averages are misleading without trace-level data.

4 Subjective Quality Signals

"Was the answer helpful?" is hard to measure. You need automated evaluators aligned with human judgment.



Five Pillars of AgentOps

A complete framework for production agent systems

1 Tracing & Observability

Capture inputs, outputs, tool calls, latency, and token usage for every agent invocation.

2 Evaluation

Automated quality scoring with LLM-as-judge: correctness, faithfulness, and custom metrics.

3 Human Feedback

Collect user signals – thumbs up/down, corrections, annotations – to ground automated scores in reality.

4 Production Monitoring

Track quality, latency, and cost in production. Detect drift and alert on regressions before users notice.

5 Governance & Lifecycle

Version models, enforce eval gates in CI/CD, manage deployment stages, and maintain audit trails.



Key Takeaways

1 Agents need the five pillars of agent ops. Ad-hoc monitoring doesn't scale — you need tracing, evaluation, human feedback, production monitoring, and governance from the start.

2 MLflow is the connective tissue. One API for tracing, evaluation, and lifecycle management — whether on Databricks or open-source.

3 Human feedback and production monitoring close the loop. Automated scores are essential, but user signals and live metrics ensure quality in the real world.

4 Governance enables confident shipping. Version models, enforce eval gates in CI/CD, and maintain audit trails. The teams that iterate fastest measure everything.



Resources & Questions

**Databricks is
Hiring!**



grnh.se/28f389a72us

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GitHub



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Add me on LinkedIn!

stephanie@databricks.com

